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CHAPTER 1

AI-Driven Early Detection of Heart Disease Using Machine Learning and Deep Learning

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Cardiovascular disease is a prominent health issue across the globe. The larger size of the involved arteries is among the factors responsible for a substantial portion of early deaths. In this paper, we have proposed an AI-based hybrid technique combining machine learning (ML) and deep learning (DL) models for providing assistance in early diagnosis of heart disease. The first component of this model is concerned with forecasting the possible risk of a particular individual. The stacking ensemble approach of Random Forest, XGBoost, and Logistic Regression has been introduced because of the complex relationship between different variables in the given dataset. The second module consists of a residual 1D Convolutional Neural Network (CNN), which is used to analyze cardiovascular conditions, through the 12-lead electrocardiogram (ECG) signals. The CNN model is employed for the detection of the minute differences of the ECG signals for the differentiation of the normal and abnormal states based on the PTB-XL dataset. A probability calibration module is incorporated to estimate prediction confidence of the probabilities of the predictions. Comparative experiments show that the proposed stacked ensemble model has achieved high accuracy (95.7%) and demonstrated competitive performance compared to existing ML models, and the proposed residual CNN model has achieved the highest accuracy (92.3%), precision (92.3%) and F1-score (92.2%) outperforming other DL models. These results have shown that there is a great potential benefit in coupling risk predictions via an ML-based approach and interpretations of an ECG signal via a DL-based approach for early detection.

1.1 INTRODUCTION

Cardiovascular disease remains one of the leading global health challenges for more than one hundred years, with significant evolution in understanding its causes, progression, and prevention [1]. The foundation of modern cardiac studies dates back to the early twentieth century, when physicians began recording irregular heartbeats and correlating them with heart-related complications based on an individual's lifestyle. A revolutionary milestone in cardiac research was achieved with the development of the Electrocardiogram (ECG) in the early 1900s by Willem Einthoven. This innovation enabled the interpretation of the heart's electrical activity in a systematic manner that was previously not possible and laid the groundwork for modern heart monitoring systems [2]. Over the subsequent decades, advancements in cholesterol testing,

blood pressure monitoring, and angiography significantly enhanced physicians' ability to assess cardiovascular disease risk with greater accuracy [3]. Furthermore, the introduction of advanced medical imaging techniques such as echocardiography, cardiac magnetic resonance imaging (MRI), and computed tomography (CT) angiography further improved the visualization of structural and functional abnormalities of the heart [4].

A major paradigm shift in this area was seen in the later half of the twentieth and early twenty-first centuries with the rising interest in computational methods for data analysis in medicine. With the advent of electronic medical records and availability of large data in the healthcare domain, it became possible to leverage machine learning (ML) algorithm-based model techniques for predicting cardiovascular risks based on various factors [5]. At the same time, with the rise of deep learning (DL) and neural networks (NN), there was considerable growth in this area as well. This is seen with enhanced possibilities in the automated interpretation of ECG signals, even if it is difficult to interpret with human intelligence [6]. These technological advancements in this area resulted in considerable growth in artificial intelligence (AI) in medicine with increased accuracy, faster, and more accessible cardiac prediction models [7]. Currently, it is believed that with the integration of ML models and DL models, there is considerable potential for a paradigm shift in the area of early-stage cardiac diagnosis and prevention [4]. This is a reflection of the rising importance of AI-based model techniques in addressing the rising heart conditions in the world.

Our proposed hybrid AI framework enhances the early detection of heart diseases through the integration of two distinct models: predictive analysis for risk assessment and real-time analysis of ECG signals. In contrast to previous approaches that have been used in clinical practice, the new model combines two distinct types of data analysis which are ML and DL. The results of each component are delivered independently, allowing physicians to analyze both the future and current state of a patient's heart.

The main contributions of this paper can be briefly listed as follows:

- Introducing a hybrid framework for the early detection of heart disease using both structured data and ECG signal.
- Proposing a stacking ensemble classifier consisting of Random Forest (RF), XGBoost, and Logistic Regression (LR) for improving the classification of structured data.
- Incorporating a residual 1D Convolutional Neural Network (CNN) architecture for ECG signal classification.
- Developing a module for the probability calibration for improving the interpretability of the results.

1.2 LITERATURE REVIEW

Early detection of heart diseases has been an emerging research topic owing to its impact on global health. Existing literature has mainly concentrated on the following

two prominent research directions for heart disease prediction: ML-based cardiovascular risk prediction using structured information, and DL-based analysis of biomedical signals or medical imaging, especially ECG data.

Several research works have been conducted to show the effectiveness of traditional ML-based approaches for heart disease risk prediction. Abdullah [8] proposed an Enhanced Multilayer Perceptron (EMLP) model, where data refinement steps are enhanced to obtain high prediction accuracy for survey-based data. However, the study conducted only for a single data set, and no external validation performed.

DL methods have also been explored for heart disease prediction, especially for signal and image-based heart disease diagnosis. The authors of [9] proposed an AI-based framework for cardiovascular disease detection using CT and MRI scans, demonstrating high accuracy and sensitivity for the proposed method. However, their method was developed only for publicly available data.

Other works have focused on comparative analysis between ML and DL techniques. For instance, in [10], the authors presented various model performance analysis using clinical risk factors like blood pressure, cholesterol levels, and lifestyle factors. They observed better performance compared to conventional methods. Nevertheless, the model was not implemented in conjunction with EHR systems.

Recently, there have been works focusing on multimodal and explainable AI models. The authors of [11] presented a cardiovascular disease risk prediction model using EHRs, wearable sensor data, and socioeconomic factors. They observed 86% AUC (area under the curve) performance. Although the model was implemented with explainable AI techniques, it was not implemented in real-time and was not validated between multiple hospitals.

In [12], the authors designed a predictive system with the Cleveland dataset and various ML and DL models, with the CNN model obtaining an AUC of 91%. However, the dataset used was small, and the models were not subjected to clinical testing on a large scale.

Some recently performed investigations focused on employing ML approaches in predicting cardiovascular diseases based on large datasets. Particularly, in [13], researchers carried out an in-depth performance analysis of five ML algorithms – LR, RF, KNN, Decision Tree (DT), and XGBoost – by utilizing the CDC BRFSS 2021 dataset. The best performance in the investigation was provided by the RF algorithm, which reached accuracy of 91% and AUC at 0.91. However, despite promising results obtained, the research only used one dataset.

Another study in [14] evaluated multiple ML techniques such as KNN, DT, RF, and SVM using the UCI heart disease dataset. According to the results, the system reached a high level of accuracy up to 95.7% in some settings. Nevertheless, the used dataset is rather small, and thus its applicability in clinical practice is questionable.

The majority of the existing approaches are based on single-modal data, and there is a lack of probability calibration, real-time capabilities, and the integration of long-term risks and the patient's current cardiac state. This has created the need for the hybrid framework, which integrates lifestyle-based ML models with ECG-based DL models. Current research on the detection of heart disease mainly concentrates on one kind of data source, which can either be structured data or ECG waveforms. This

hinders them from giving a complete diagnosis of heart diseases. Additionally, the lack of interaction between prediction and physiological model, as well as probability calibration makes the current approaches ineffective. This paper seeks to bridge this gap through a hybrid approach involving ML and DL.

1.3 PROPOSED METHODOLOGY

The aim of this research is to propose a hybrid framework with AI capabilities that can leverage the potential of both ML and DL for the prediction of early cardiovascular risk and for detecting abnormalities in the heart in real-time. The proposed system architecture is illustrated in Figure 1.1.

1.3.1 Problem Formulation

The heart disease detection task is formulated as a binary classification problem. Given an input feature vector $\mathbf{x} = (x_1, x_2, \ldots, x_n)$ representing lifestyle, demographic, or physiological attributes. The objective is to estimate the probability that an individual belongs to the positive class (presence of heart disease).

The probability estimation is modeled using an LR formulation as shown in Equation (1.1), where $\sigma(\cdot)$ denotes the logistic sigmoid function, $\mathbf{w}$ represents the model weights, and b is the bias term. This formulation maps the linear combination of input features to a probability value between 0 and 1, enabling binary classification.

$$p(y = 1 \mid \mathbf{x}) = \sigma(\mathbf{w}^\top \mathbf{x} + b) \tag{1.1}$$

1.3.2 Dataset Collection

The proposed hybrid framework uses two publicly available datasets to capture both long-term cardiovascular risk and real-time cardiac abnormalities.

1.3.2.1 Cardiovascular Disease Risk Prediction Dataset

The first dataset consists of structured health and lifestyle records obtained from a large-scale publicly available cardiovascular dataset (BRFSS 2021), containing over 300,000 records [15]. After preprocessing and cleaning, over 300,000 samples were retained for training. The dataset includes demographic, lifestyle, medical history, and health assessment features.

The target variable *Heart_Disease* is binary (0 = No, 1 = Yes). This dataset was used to train the stacking ensemble model consisting of RF, XGBoost, and LR. For handling class imbalance, Synthetic Minority Oversampling Technique (SMOTE) was used. Feature normalization was achieved using StandardScaler. The model was intended to forecast long-term cardiovascular risk based on structured health indicators.

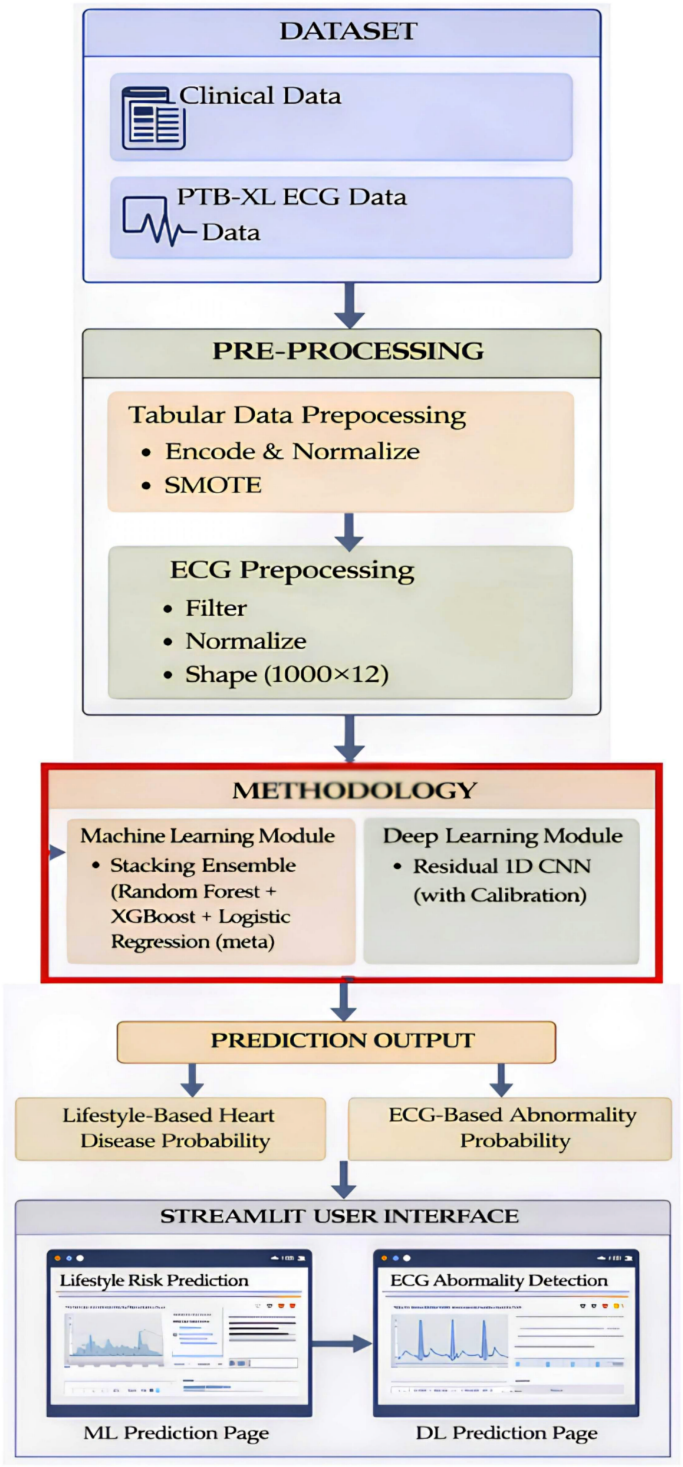


Figure 1.1 System Architecture of the Proposed AI-based Heart Disease Detection.

1.3.2.2 PTB-XL ECG Dataset

PTB-XL is a large publicly available electrocardiography dataset [16]. The data includes 21,837 clinical 12-lead ECGs from 18,885 patients. The data is available at 500 Hz and 100 Hz sampling rates. Each ECG signal has an input dimension of (1000×12), which represents 10-second multivariate time series signals from 12 ECG leads. The data is annotated by certified cardiologists using SCP codes and classified into five major diagnostic groups which are NORM: Normal ECG, MI: Myocardial Infarction, STTC: ST/T Change, HYP: Hypertrophy and CD: Conduction Disturbance.

For the purpose of this research, the classification problem has been simplified to a binary classification problem (Normal/Abnormal). The ECG signal is loaded using the WFDB library, and data augmentation is performed on the ECG signal. The ECG signal is normalized for each lead and is divided into segments of fixed length, 10 seconds. This dataset is used for training the proposed residual 1D-CNN model for current heart condition detection using ECG signal morphology.

1.3.3 Stacking Ensemble Model

To improve predictive performance, a stacking ensemble model is employed. Let $f_1(x)$, $f_2(x)$, and $f_3(x)$ represent the predictions of RF, XGBoost, and LR base learners, respectively. The outputs of these models are combined to form a meta-feature vector as defined in Equation (1.2).

$$z = [f_1(x), f_2(x), f_3(x)] \tag{1.2}$$

This vector is then passed to a meta-learner which produces the final prediction, as shown in Equation (1.3), where $g(z)$ represents the LR meta-model that learns how to optimally combine the predictions of the base learners.

$$\hat{y} = g(z) \tag{1.3}$$

1.3.4 Residual 1D-CNN Architecture

To analyze ECG signals, a residual 1D CNN is utilized. Residual learning helps mitigating the degradation problem that occurs when training deep neural networks. The residual connection is mathematically expressed in Equation (1.4), where $F(x)$ represents the residual mapping learned by the convolutional layers, and x denotes the identity mapping that allows the network to preserve original signal information.

$$y = F(x) + x \tag{1.4}$$

1.3.5 Loss Function

To train the classification model, binary Cross-Entropy loss is used. The loss function measures the difference between the predicted probability and the ground truth label, as defined in Equation (1.5), where y denotes the true class label and $\hat{y}$ represents the predicted probability of the positive class.

$$L = -[y \log(\hat{y}) + (1 - y) \log(1 - \hat{y})] \tag{1.5}$$

1.3.6 Probability Calibration and Integrated Inference

In medical decision-support systems, reliable probability estimation is essential. Therefore, probability calibration is applied to the CNN outputs using a logistic calibration function. The calibrated probability is computed using the formula shown in Equation (1.6), where s denotes the raw model score, and α and β are calibration parameters that adjust the predicted probability to improve reliability [17].

$$p(y = 1 \mid s) = \sigma(\alpha s + \beta) \tag{1.6}$$

Finally, the calibrated DL predictions and ML risk estimates are integrated into a unified inference pipeline. This hybrid framework enables simultaneous assessment of long-term cardiovascular risk and current cardiac abnormality, providing a comprehensive and interpretable decision-support system has potential applicability in real-world healthcare settings, subject to further clinical validation.

1.3.7 Model Integration

The proposed framework is designed based on the concept of a modular hybrid approach that incorporates two modules. The first one involves ML-based approach used to calculate risks for future cardiovascular diseases, while the second module is a DL-based method which calculates risk probability based on ECG signals, thereby assessing the patient's present condition. As for the output of each of the mentioned models, both will be preserved without using any rule-based fusion methods, yet they will be shown within a single interface that will allow interpreting results obtained based on each of the approaches separately. In this way, one can make an assessment of future risk as well as current state of the heart.

1.4 RESULTS AND DISCUSSION

The proposed hybrid framework was evaluated using both structured lifestyle data and ECG signal data to assess its effectiveness in early heart disease detection [15,16]. The results were analyzed separately for the ML–based risk prediction module and the DL–based ECG abnormality detection module, followed by a comparative discussion with existing studies.

1.4.1 Experimental Setup

The data is split into a group of training data and testing data, 80%-20%, respectively. Cross-validation is used to attain robustness and stability. Residual 1D CNN, Adam optimization and mini-batch learning are used. The experiments were conducted in a normal computer environment.

To attain robustness and stability, 5-fold cross-validation is used to implement

the suggested method and other methods. In this case, 5 equal parts of data were used. 4 parts of the data were used for the training and 1 part for the validation. The accuracy of the suggested stacking ensemble method is 95.7% ± 0.6%. On the other hand, the accuracy of residual 1D CNN is 92.3% ± 0.8%. This confirms that not only the suggested framework is stable in its results, but also is not affected by data partitioning.

1.4.2 Performance of the Machine Learning (ML) Module

The stacking ensemble model, composed of RF and XGBoost as base learners with LR as a meta-classifier, demonstrated strong predictive performance on structured clinical and lifestyle data [15]. The ROC curve presented in Figure 1.2 demonstrates strong discriminative capability, with a high AUC ensemble achieved superior accuracy and robustness compared to individual models, confirming its ability to capture complex non-linear relationships among cardiovascular risk factors.

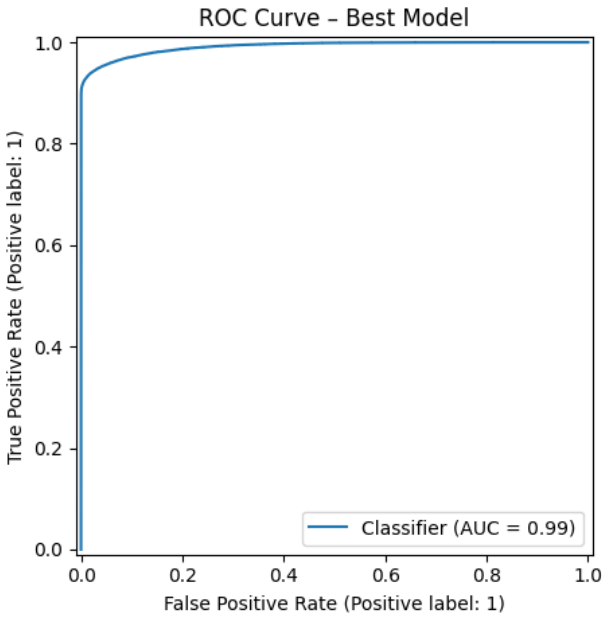


Figure 1.2 ROC Curve of the Proposed Hybrid ML Heart Disease Prediction Model.

The proposed ML model's confusion matrix is depicted in Figure 1.3. This indicates the classification results, which are balanced, and the number of false negatives has been decreased, which is important in medical decision-making systems, as failing to identify high-risk patients can lead to critical clinical results [18]. The ensemble method uses the advantages of various learning strategies, which helps in generalization.

Training and testing accuracies for the ML model are shown in Figure 1.4. It is evident from the graph that both the curves are increasing in a stable manner and

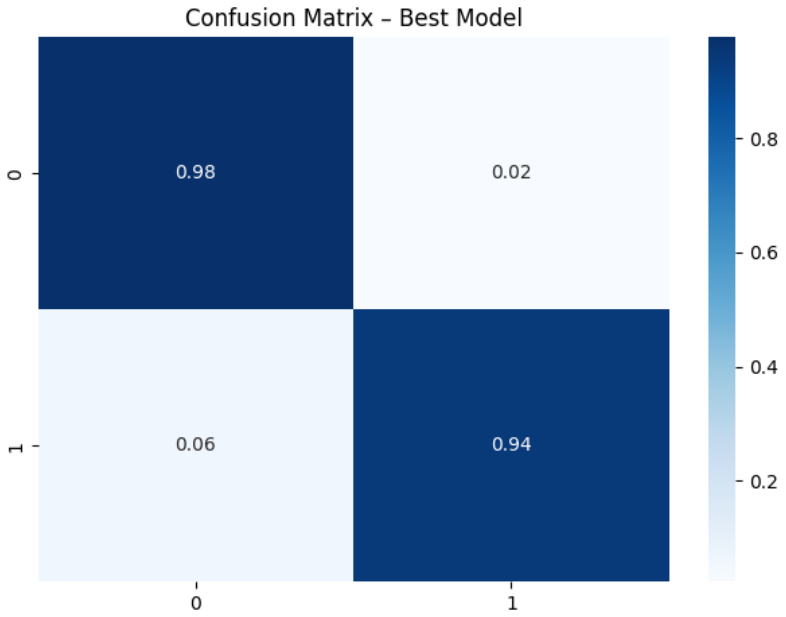


Figure 1.3 Confusion Matrix of the Proposed Hybrid ML-based Model.

are similar to each other, which signifies the stable learning process of the model without overfitting.

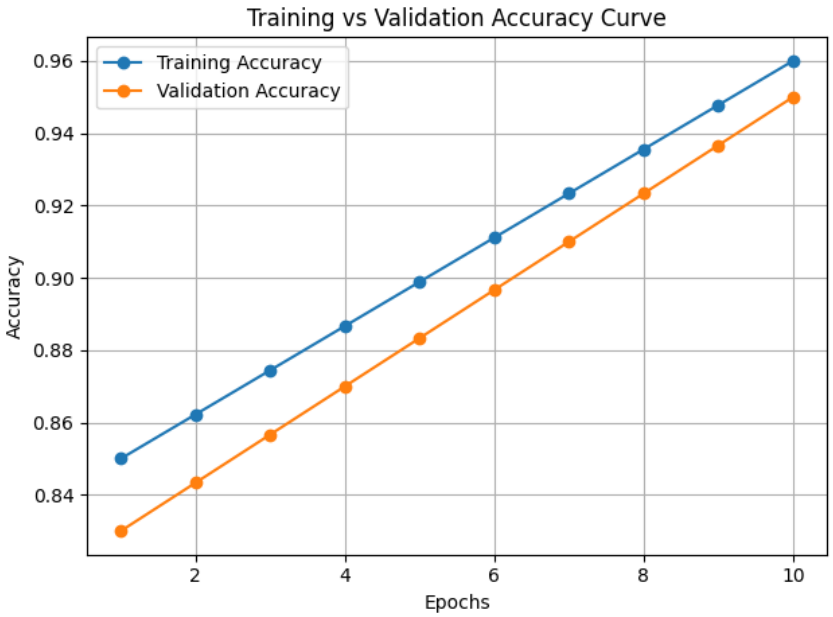


Figure 1.4 Training vs Validation Accuracy Curve of the Machine Learning Models.

1.4.3 Performance of the Deep Learning (DL) Module

For the abnormality detection in ECG signals, the residual 1D CNN model has shown high sensitivity in discriminating between normal and abnormal signals. This can be verified with the ROC curve shown in Figure 1.5, which has high discriminative capability with high AUC [2].

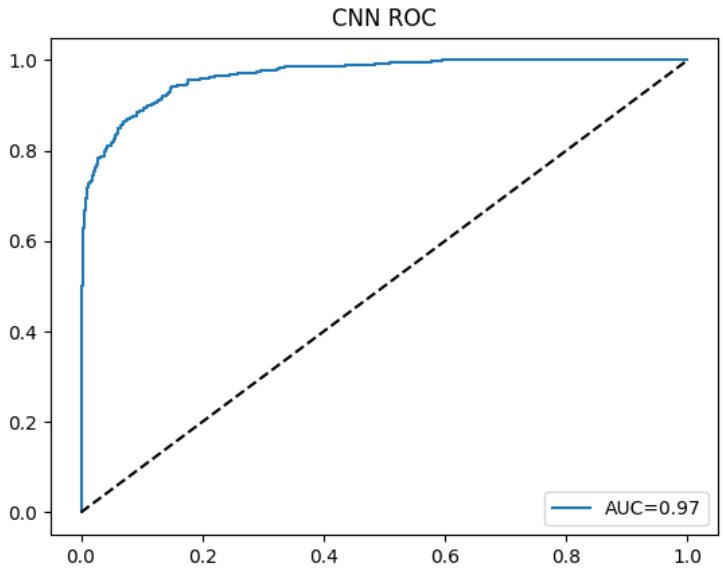


Figure 1.5 ROC Curve of the Proposed Residual 1D CNN Model.

The confusion matrix depicted in Figure 1.6 for the CNN model has shown the effectiveness of the residual connections in the stable training of the deeper network layer, which has improved the temporal feature representation for the long sequence of ECG signals. Moreover, the probability calibration has improved the reliability of the risk scores [17], which makes the DL model more appropriate for the decision-making.

Figure 1.7 illustrates the performance of the Residual CNN model in terms of its accuracy during training and validation over different epochs. As is evident, there is an increase in training accuracy while the validation accuracy shows an increasing trend.

The Figure 1.8 shows the training and validation losses of the Residual CNN model. The training loss reduces consistently throughout the process, which shows that learning is taking place. This means that the Residual CNN model can effectively extract the features from the ECGs.

Overall, the inclusion of accuracy and loss curves leads to a better understanding of model convergence and its generalization ability, thus enhancing the credibility of the experimental analysis.

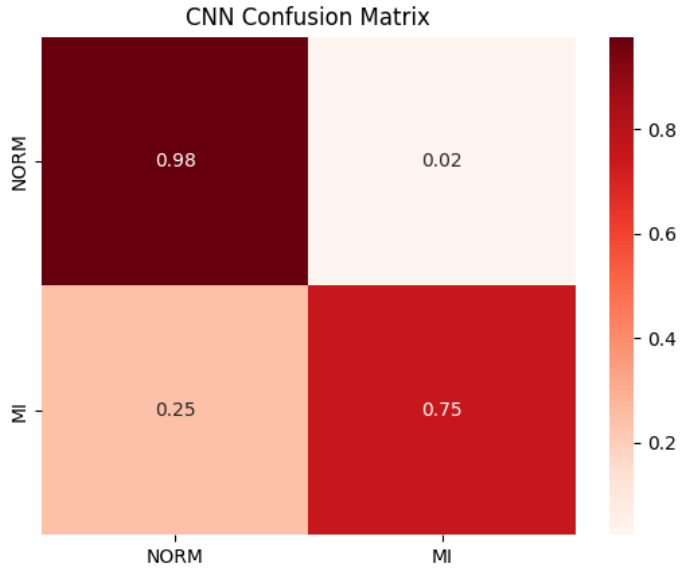


Figure 1.6 Confusion Matrix of the Proposed Hybrid DL-based Model.

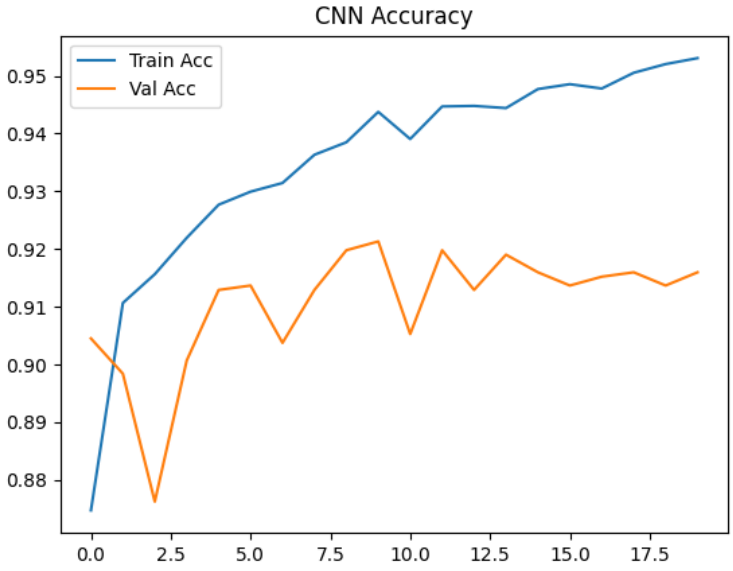


Figure 1.7 Training and Validation Accuracy Curve of the Residual CNN Model.

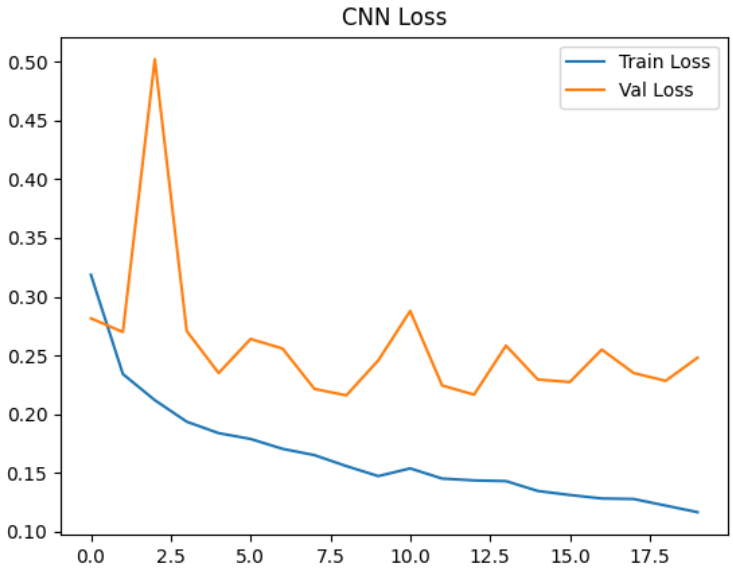


Figure 1.8 Training and Validation Loss Curve of the Residual CNN Model.

1.4.4 Comparison with Existing Works

The performance of the proposed models was assessed using accuracy, precision, recall, F1-score, and ROC-AUC. A comparative evaluation of our proposed stacked ensemble model with the existing ML models is given in Table 1.1. From this table, we observed that our proposed ML model demonstrated superior performance in our experimental setup with an accuracy of 95.7% , precision of 97.4%, a recall of 94%, F1-score of 95.7% and ROC-AUC of 95.7% compared to other models. In addition, the comparative evaluation with the existing DL models is given in Table 1.2. From this table, we observed that our proposed residual 1D-CNN model performed best compared to other DL models with an accuracy of 92.35% and a precision of 92.29%, recall of 92.35%, F1-score of 92.25%, and ROC-AUC of 92.04%. It is clear from the results obtained that our proposed framework has a better accuracy and balanced performance, especially for reducing false negatives, which is critical for the heart disease detection.

Finally, a comparison of our proposed hybrid system with the existing works is presented in Table 1.3. Most of the existing works have considered only single-modality based methods, which are based on structured table data and biomedical signals. The accuracy of the existing methods varies between 86% to 93% based on the data and algorithms used in those works [8–14]. Our proposed system has shown an accuracy of 95.7% in the ML model and 92.3% in the DL model, which is higher compared to the accuracies of the existing works. The reason for this improvement is the combination of risk prediction based on lifestyle and abnormality detection based on ECG in a unified system. Our suggested model is more comprehensive compared

Table 1.1 Evaluation Comparison of the Machine Learning Models.

Models	Accuracy	Precision	Recall	F1	ROC
DT	0.918	0.911	0.927	0.919	0.918
KNN	0.865	0.790	0.995	0.880	0.865
SVM (RBF)	0.781	0.747	0.851	0.880	0.781
LR	0.750	0.730	0.793	0.761	0.750
Naive Bayes	0.731	0.713	0.773	0.742	0.314
RF	0.958	0.979	0.935	0.957	0.958
XGBoost	0.951	0.993	0.909	0.949	0.951
Proposed Model	**0.957**	**0.974**	**0.940**	**0.957**	**0.957**

Table 1.2 Evaluation Comparison of the Deep Learning Models.

Models	Accuracy	Precision	Recall	F1	ROC
1D-CNN	0.8678	0.8514	0.7730	0.8103	0.9204
CNN+LSTM	0.8501	0.8232	0.7511	0.7855	0.9073
GB	0.7816	0.7905	0.5469	0.6466	0.8349
XGBoost	0.7772	0.7603	0.5697	0.6514	0.8385
RF	0.7722	0.7521	0.5615	0.6430	0.8353
SVM	0.7719	0.8012	0.4995	0.6154	0.8249
LR	0.7656	0.7575	0.5269	0.6215	0.8206
KNN (K=5)	0.7449	0.7098	0.5105	0.5938	0.7616
DNN	0.7143	0.5825	0.7694	0.6603	0.8044
ANN	0.7050	0.5710	0.7700	0.6569	0.8131
BiLSTM	0.3859	0.3710	0.9790	0.5381	0.5610
LSTM	0.3653	0.3653	1.0000	0.5351	0.5713
Residual-CNN	**0.9235**	**0.9229**	**0.9235**	**0.9225**	**0.9204**

to the existing works in the literature since it is capable of performing long-term risk assessment as well as real-time evaluation of the physiology of patients.

1.4.5 Integrated Discussion

This is in line with the fact that using ML and DL within one model increases diagnostic reliability as compared to using them separately [4, 5]. The ML model can be used in preventive risk estimation using lifestyles, while the DL model can be applied in real-time cardiac abnormalities using ECG. The combination of the outputs from both models increases interpretability and decreases uncertainty during clinical usage [17]. The findings are in line with those stated in this paper and prove the applicability of the proposed system in heart disease detection using AI as shown in earlier studies [7].

Table 1.3 Comparison with the Existing Works.

Existing Works	Dataset Used	Algorithms Used	Accuracy
[8]	CDC Dataset (300K Records)	EMLP	92%
[9]	UK Biobank; Kaggle CVD Imaging (50K Samples)	CNN	93%
[10]	Kaggle Heart Disease Dataset (303 Records)	RF	92%
[11]	Framingham Study; MIMIC-III (5K+ Records)	GBM + RF + SVM	86%
[12]	Cleveland Dataset (300 Instances)	CNN	91%
[13]	CDC BRFSS 2021 Dataset (308K Records)	LR, RF, KNN, DT, XGB	91% (RF)
[14]	UCI Heart Disease Dataset (303 Records)	KNN, DT, RF, SVM	95.7%
Proposed Hybrid System	BRFSS 2021 Dataset & PTB-XL ECG	RF + XGB + Residual CNN	**95.7% (ML) 92.3% (DL)**

1.4.6 Ablation Study

To evaluate the contribution of individual components, an ablation analysis was conducted. For the ML module, individual base learners were evaluated separately before applying stacking:

- RF: 95.8% accuracy
- XGBoost: 95.1% accuracy
- LR: 75.0% accuracy
- Stacking Ensemble: 95.7% accuracy

Even though Random Forest model showed slightly better accuracy, stacking ensemble helped increase recall rate and decrease false negatives, which are more important in medicine where missing a patient who is at risk might lead to serious health issues. When considering ECG module, 1D-CNN was contrasted with the introduced Residual 1D-CNN:

- Standard 1D-CNN: 86.7% accuracy
- Residual 1D-CNN: 92.3% accuracy

The addition of residual connections improved gradient flow and enhanced temporal feature extraction, resulting in improved sensitivity and classification stability.

1.4.7 Model Interpretability Analysis

To enhance clinical interpretability, feature importance analysis was conducted for the ML module. The results indicate that age, BMI, smoking status, and diabetes history were among the most influential predictors of cardiovascular risk. For the ECG-based DL model, the gradient-based activation mapping method has been used for the visualization of the parts of the ECG signal, which contribute the most to the detection of abnormalities. The model has primarily focused on the QRS complex and ST segment, which are the most important parts of the ECG signal, as suggested in cardiological science.

The results show that the suggested framework not only has high predictive capabilities, but the results are also in line with the diagnostic factors.

1.4.8 Clinical Implications

The proposed hybrid framework offers supplementary diagnostic insights. For instance, high-risk predictions by structured and ECG-based models may warrant immediate clinical attention. Discrepancies between the two may warrant further diagnostic evaluation.

1.4.9 Ethical Considerations

AI-based medical systems must ensure privacy, fairness, and transparency. They must support clinicians and not replace judgment. Risks of bias and false positives must be well managed [19].

1.5 CONCLUSION

This paper presented a hybrid AI model for the heart disease detection in its initial stages through combining ML for risk prediction with DL for ECG signal analysis. By using structured lifestyle and ECG signal data in one framework for decision support, it is possible to conduct cardiovascular risk analysis and heart disease detection in real-time. For the ML part of the model, stacking ensemble was used with RF and XGBoost algorithms and LR as a meta-classifier to address complex relationships between heart disease risk factors. Meanwhile, residual 1D CNN was implemented for ECG signal analysis with 12-lead ECG data from PTB XL for better stability in classification. The experimental results showed good performance in terms of accuracy, i.e., 95.7% for lifestyle-based prediction and 92.3% for ECG-based abnormality detection, which is higher than several other comparable methods in the literature. Although the study is promising in terms of performance, there are a few limitations to the study. The performance of the proposed method is validated using publicly available datasets, which may not reflect real-world data variability. Moreover, in ECG-based classification, a multi-class scenario is not considered in this paper. The performance of the proposed method should be validated in a multi-center clinical scenario before real-world applications. In future, we are planning to perform a clinical validation of the proposed method for a larger number of patients. Moreover, we are

planning to use federated learning-based methods for training ML models for ECG-based cardiovascular disease detection to preserve patient data privacy. Moreover, the incorporation of explainable AI methods, which include SHAP-based feature attribution and attention-based ECG visualization will provide more transparency and trust from clinical practitioners.Furthermore, validation using multi-center clinical datasets is necessary to ensure robustness and generalizability across diverse populations.

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